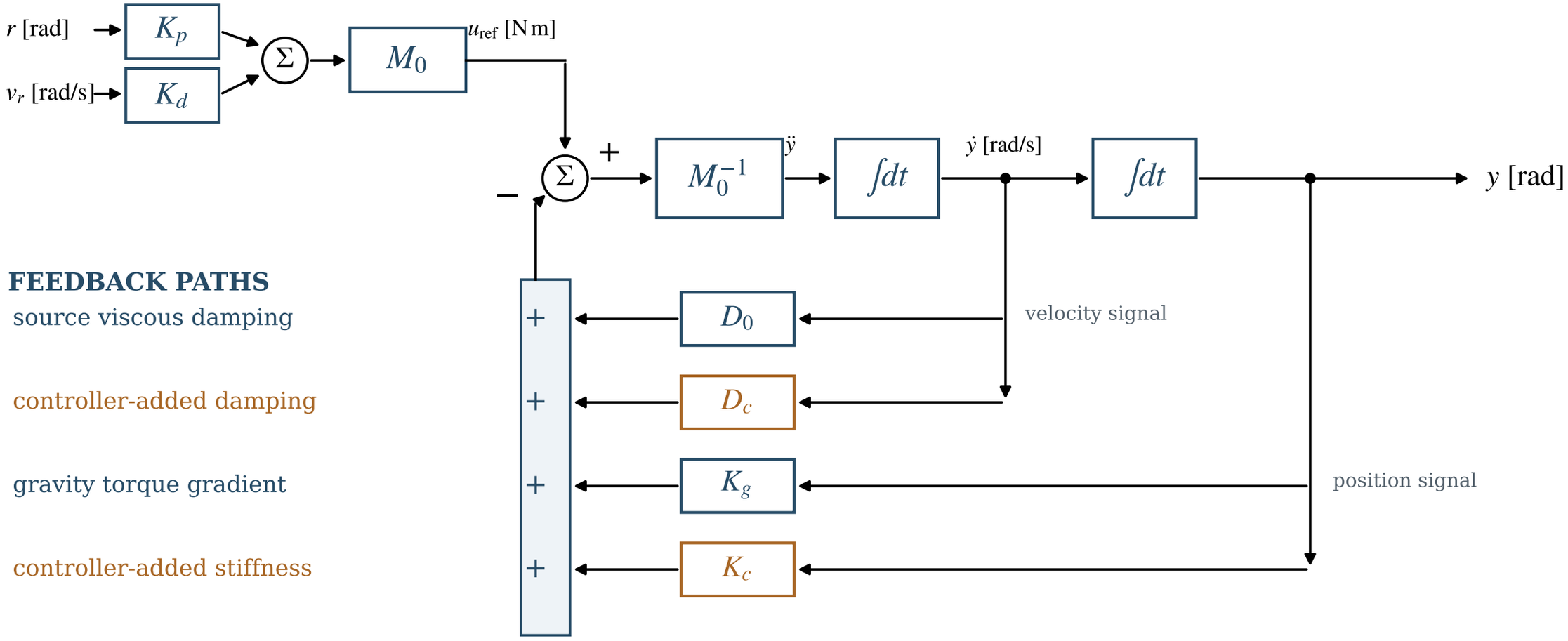
V16 | Nominal 12-joint equivalence — analysis only

Fixed-base, no-contact source model; not the runtime WBC. Blue gains are physical, orange gains are ideal equivalent control terms.

REFERENCE TORQUE



COEFFICIENTS WITH PHYSICAL UNITS

$$D_c = M_0 K_d - D_0$$

$$K_c = M_0(1+\alpha)K_p - K_g$$

$$D_{\text{eff}} = D_0 + D_c$$

$$K_{\text{eff}} = K_g + K_c$$

D [N m s/rad], K [N m/rad]. Source D0 = 1.5 I; actual USD damping = 0. This nominal equivalence is not the runtime WBC on page 1.